Journal of King Saud University Computer and Information Sciences (2025) 37:282
https://doi.org/10.1007/s44443-025-00312-y

ORIGINAL PAPER

SPEAR-net: a neuro-inspired causal perception and episodic memory framework for fine-grained and context-aware soccer action recognition

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Received: 8 May 2025 / Accepted: 24 September 2025 / Published online: 28 October 2025

Abstract

Accurate and context-aware action recognition in soccer remains a challenging task due to the sport's dynamic nature, overlapping actions, and complex causal relationships among events. Traditional video understanding models often rely on dense optical flow and static frame-based processing, which struggle in scenarios involving occlusion, background clutter, or subtle gameplay nuances. Furthermore, many existing systems fail to incorporate temporal reasoning or causal inference, leading to suboptimal recognition of interdependent actions like tackles, fouls, or passes. To address these limitations. This research aims to develop a cognitively inspired framework that overcomes the limitations of traditional video understanding models, particularly their reliance on dense optical flow and inability to model temporal causality and context effectively. We propose SPEAR-Net (Spiking-Perception and Episodic Abstraction for Recognition Network), a neuro-inspired action recognition system that emulates brain-like perception and reasoning. It introduces modules such as the Neuro-Perception Encoding Module (NPEM) for sparse event encoding, the Causal Relational Decoder (CRD) for DAG-based causal reasoning, the Context-Aware Environment Encoder (CAEE) for semantic context embedding, and the Micro–Macro Event Alignment (MMEA) for hierarchical temporal abstraction. Reliability is further enhanced through the Episodic Memory Replay Module (EMRM) and Self-Critic Refinement Agent (SCRA) for memory-based adaptation and confidence calibration. Empirical results on the SoccerNet-v2 benchmark demonstrate that SPEAR-Net outperforms state-of-the-art models with a Top-1 accuracy of 89.6%, robustness under occlusion (only 4.5% accuracy drop), and significantly reduced misclassification and uncertainty post-refinement, making it highly suitable for real-time, high-stakes sports analytics.

Keywords Soccer action recognition · Context-aware recognition · Causal reasoning · Event alignment · Neuro-perception encoding · Self-critic network

1 Introduction

Substantial work in automated event analysis and temporal action interpretation has been motivated by the explosive growth in sports video viewing, especially in sports such as soccer (Akan and Varlı 2023). A challenging testbed for fine-grained video analysis, including challenges such as event classification, temporal localization, and action detection, is offered by the fast pace of soccer and its unstructured playing patterns (Seweryn et al. 2023). Due to the inherent difficulties presented by occlusions, rapid changes, multiple views, and low inter-class variability, these tasks provide benchmarks for the broader action recognition community as well as enable applications in sports analytics, coaching, and broadcasting (Zhang et al. 2022a). One of the primary challenges in soccer video analysis is one of the fine and sometimes ambiguous differences between apparently similar activities (Fele and Campagnolo 2024). For instance, a "goal celebration" and a "team huddle" might share the same temporal and spatial features, but in a high-impact event, e.g., foul or penalty, camera jitter or sudden occlusions might mask critical motion signals (Gong et al. 2021). The performance of purely spatial models is also hindered by the dynamic camera pan and cut-frequent shot changes that occur in broadcast-type recordings. These challenges highlight the importance of robust spatiotemporal modelling that

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can both process existing frames in an efficient manner and adapt over time based on contextual cues. For extracting spatial features from single frames or short clips, conventional CNN-based architectures have shown decent performance (Suresha et al. 2020) (Aldahoul et al. 2022).

However, they often do not distinguish between context-dependent activity patterns and store long-range temporal relationships (Kim et al. 2022). Recurrent units or 3D convolutional networks have been employed in several studies to address temporal dependencies (Xie et al. 2020). Although 3D CNNs like I3D or R (2 + 1) D can model short-term temporal properties, they struggle to remember longer sequences, particularly in sports scenarios where there are limited events (Sun et al. 2023). Even though recurrent-based models are theoretically suitable for sequence modelling, they often have difficulty with vanishing gradients and memory access problems, which restricts their ability to leverage episodic signals from past game instances. Transformer-based architectures have gained popularity in recent times due to their remarkable capacity to model global dependencies over time (Sajun et al. 2024). Activities in long films can be effectively localized with attention-based approaches, which is evidenced by such works as SpotFormer, TMSR, and TAS-SV (Yang et al. 2022). These models can reason throughout the entire match since they attend to temporally disperse stimuli (Henson et al. 2024). But they must pay a computational cost for their worldwide attention, and their static memory encoding often limits their ability to adapt to partially visible or ambiguous scenes (Xie et al. 2022). In addition, most transformer models assume that context is always relevant at all points in time, which is not necessarily true in the fluid dynamics of soccer, where tactical or circumstantial context affects the interpretation of previous actions in the current moment.

Adding dynamic memory structures to the model design is a new temporal video understanding direction that seeks to address these limitations (Li et al. 2024). Models can make more precise predictions by having memory modules that selectively retain and reuse contextual patterns of high value, particularly when there is visual degradation such as blur, occlusion, or disconnected viewpoints. This is in line with cognitive strategies in human perception, wherein experience and episodic memory are crucial for judging rapidly in the presence of insufficient information or distinguishing ambiguous situations. Construction of models that can selectively adjust their conclusions based on historical data and enhanced temporal reasoning in the presence of ambiguity are the pillars of our work in this direction. Our approach is consistent with recent advances in hybrid architectures that integrate memory-driven feedback and spatiotemporal encoders. There is a lack of integrated strategies that adapt dynamically to real-time situations and include self-improvement logic for building confidence and decision stability, although existing works have explored these in isolation, e.g., action spotting using transformers or graph networks representing player dynamics. Therefore, three main research gaps are the driving forces of this work:

1. *Lack of robust handling of ambiguous scenarios* – Modern models often struggle with scenarios involving overlapping player shapes, visual clutter, or occlusions.
2. *Absence of dynamic memory utilization* – Even while certain methods use fixed attention over sequences, they cannot give uniform decision-making because of episodic memory.
3. *Limited refinement mechanisms post-inference* – Post-decision adjustment through internal feedback or confidence calibration is facilitated by relatively few systems today.

Our evaluation framework is geared towards encouraging models with context-awareness, high temporal alignment, and adaptive reasoning under uncertainty to address these problems. Employing a new memory-enhanced architecture that is tested on competitive benchmarks and compared with state-of-the-art models like SpotFormer, TAS-SV, GNN-3D, and TMSR, we introduce and evaluate these concepts. This work sets the stage by establishing what the real-world problems of soccer video analysis are that need to be solved using cognitive-inspired modelling, instead of unveiling the architectural details of our model immediately. We aim to call attention to the intricate yet essential attributes that modern temporal reasoning systems must have by structuring our analysis along several sub-dimensions such as classification accuracy, robustness in ambiguity, decision refinement, handling of temporal phases, and inference-efficiency trade-offs. Within these, we have three things to offer:

- We develop an extensive test suite that simulates real soccer video analysis situations, including occlusion tests, jitter tests, and distorted sequences.
- By considering its impact on a range of measures, such as contextual accuracy, decision stability, and misclassification reduction, we propose and identify episodic memory-based reasoning as a key design direction.
- By incorporating internal refinement loops, confidence calibration methods, and adaptive feedback mechanisms that simulate human-like perceptual adaptation, we motivate future models to move beyond static inference.

In summary, our contribution is an advancement toward closing the cognitive divide in video understanding, especially in the complex, event-dense domain of soccer. We envision a next generation of video intelligence systems capable of understanding actions precisely, solidly, and transparently by developing models that not only watch and listen but also remember, compare, and learn. This paper is arranged in the format of Section II reviews related work in soccer video understanding and action recognition. Section III introduces the proposed SPEAR-Net architecture in detail, covering each cognitive-inspired module. Section IV presents experimental results and analysis, including accuracy, robustness, confidence calibration, and efficiency. Finally, Section V concludes the study.

2 Literature Survey

Cao et. al (Cao et al. 2022) introduced SpotFormer, a new transformer model designed particularly for accurate action temporal localization in soccer movies. To spot accurately, it utilizes temporal anchoring and multi-head attention. The effectiveness of the model is proved by its state-of-the-art performance on SoccerNet benchmarks. Gan et. al (Gan et al. 2022) utilized transformer models for connecting visual and audio modalities to interpret scenes in soccer video. It utilizes cross-modal attention and produces improved categorization of game phase and events. It presents a robust approach to segment semantic scenes in complex sports environments. Zhang et. al (Zhang et al. 2022b) proposed a hierarchical attention framework to emphasize significant body parts and interactions in group activity scenarios. It excels in capturing spatial–temporal relations when used with skeleton-based data. On common group activity dataset, the model outperforms existing approaches. Xu et. al (Xu 2024) categorized referee behaviour in football matches based on posture estimation and deep learning. To understand motion in the skeletal system, it builds a custom dataset and leverages graph-based models. The technology can help automate referee-event recognition. Bai et. al (Bai et al. 2022) used TransFusion a transformer-based pipeline to fuse LiDAR and camera features for accurate 3D object detection. Its robust multi-sensor fusion is not soccer-specific, but it can assist sports analytics that leverage autonomous systems. Detection accuracy and spatial reasoning are significantly enhanced by the architecture.

Roddick et. al (Roddick and Cipolla 2020) offered a hierarchical pyramid network-based method for predicting semantic occupancy maps from a single-view image. Its application is in spatial understanding, which may assist with mapping soccer fields. For real-time applications, the technique is a trade-off between speed and accuracy. Ammar et. al (Ammar et al. 2024) presented a new metaheuristic approach to global optimization problems inspired by soccer match strategy. To prevent being trapped in local optima, the algorithm mimics player tactics and dynamics. The algorithm runs competitively and has been tested on a variety of benchmarks. Montazeri et. al (Montazeri et al. 2023) introduced the Golf Optimization Algorithm (GOA), an optimization problem-solving methodology inspired by golf strategies. It is utilized within the energy field to illustrate resiliency-aware scheduling. It adds value to game-conceptual algorithm design even without any connection with soccer. Latif et. al (Latif et al. 2022) employed side data in a multitask learning setting to enhance emotion recognition. Through emotional tone analysis, the outcomes can be used to generate soccer highlights. The approach boosts robustness and generalization in audio-based tasks. Tani et. al (Tani et al. 2022) employed deep learning and domain ontologies to detect soccer offside events. Through the application of semantic modelling to rule and context definition, it enhances interpretability. The system is a step towards sports officiating with explainable AI. Baron et. al (Baron et al. 2024) proposed a large-scale underlying model trained on vast quantities of soccer video data to extend to other activities. It involves spatiotemporal hints, text, and video. The transferability to numerous downstream soccer analytics tasks is illustrated by the model. Ochin et. al (Ochin et al. 2025) simulated player interactions and detects game state changes by integrating GNNs and 3D CNNs. Through structured relational modelling, it shows improved detection of complex soccer movements. This hybrid approach excels in both spatial and temporal comprehension. Soares et. al (Soares et al. 2022) introduced temporally accurate action spotting dense anchor mechanisms. Since it identifies events at a smaller temporal granularity, it has better performance compared to typical models. With regards to pinpointing short-term soccer moves, the dense anchoring improves remembrance. Shi et. al (Shi et al. 2022) explored multi-encoder architectures to capture different visual contexts in soccer situations. Multiple viewpoints, including close-ups and full fields, are the focus of each encoder. Fusion methods supply holistic action localization for the model.

Darwish et. al (Darwish and El-Shabrway 2022) introduced STE, a spatiotemporal encoder that employs both appearance and motion signals for event detection. Its hierarchical structure effectively captures evolving game dynamics. Robust performance on SoccerNet benchmarks is shown by the results. Zhu et. al (Zhu et al. 2022) employed a standard transformer architecture to identifying and finding soccer events. Attention processes in games assist with the learning of long-term dependence. The results are not only scalable but also competitively accurate. Gautam et. al (Gautam et al. 2022) proposed video summarization of soccer games by sound intensity peaks. Notable plays are linked with fan uproar and announcer excitement. The approach is an improvement over traditional video summarizing pipelines and is lightweight. Cabado et. al (Cabado et al. 2024) evaluated the generalizability of action spotting models trained on premier league soccer to other soccer domains. It points out significant domain gaps and indicates how to fill them. The findings are critical to making sure that AI models are representative of all leagues. Mkhallati et. al (Mkhallati et al. 2023) introduced a new dataset and challenge for generating thorough, semantically dense soccer broadcast subtitles. It produces comments using language models and temporal localization. This bridges the gap between visual analysis and natural language understanding. Prat et. al [45] employed pioneering computer vision methods to lay the basis for machine sports video understanding. It classifies events in terms of scene transitions and motions. Although it is old, it still influences sports analytics studies nowadays in terms of methodology.

Research Gaps: Despite significant advancements in transformer-based and graph-driven models for soccer video analysis, critical limitations persist. Most existing approaches rely heavily on frame-based processing or static attention, lacking the ability to adaptively handle ambiguous or occluded scenarios. Models like SpotFormer and STE, while strong in temporal localization, fail to incorporate causal reasoning, limiting their interpretability. Cross-modal and multi-encoder models improve semantic richness but do not fully address temporal memory or decision refinement. Furthermore, memory usage in current systems is often fixed or shallow, preventing effective reuse of historical context in dynamic gameplay. Many methods also overlook the importance of action re-evaluation when prediction confidence is low, which reduces robustness in noisy or high-pressure match conditions. In addition, macro-level game strategies are rarely connected back to fine-grained actions, creating a disconnect in hierarchical understanding. SPEAR-Net addresses these gaps by combining neuro-inspired event encoding, causal graph reasoning, dynamic context fusion, episodic memory recall, and self-critique refinement to deliver a comprehensive, context-aware, and cognitively motivated solution.

3 SPEAR-Net Framework

We propose SPEAR-Net in Fig. 1, a neuro-inspired soccer action recognition framework that leverages brain-like processing for fine-grained and causally-aware understanding of game dynamics. SPEAR-Net is a neuro-inspired framework designed for fine-grained and context-aware soccer action recognition. Initially, with the NPEM, which converts raw video frames into sparse motion events using spiking neural encoding, eliminating the need for optical flow. CRD models cause-effect relationships between actions using DAG-based attention, enabling the system to reason why an action occurred. To ground predictions in real-world context, the CAEE and MMEA modules extract semantic scene cues and align low-level actions with high-level strategies. Finally, the EMRM and SCRA refine predictions through memory recall and confidence-based re-evaluation, ensuring robustness in ambiguous scenarios. Here, Table 1 represents the significant notations with its descriptions.

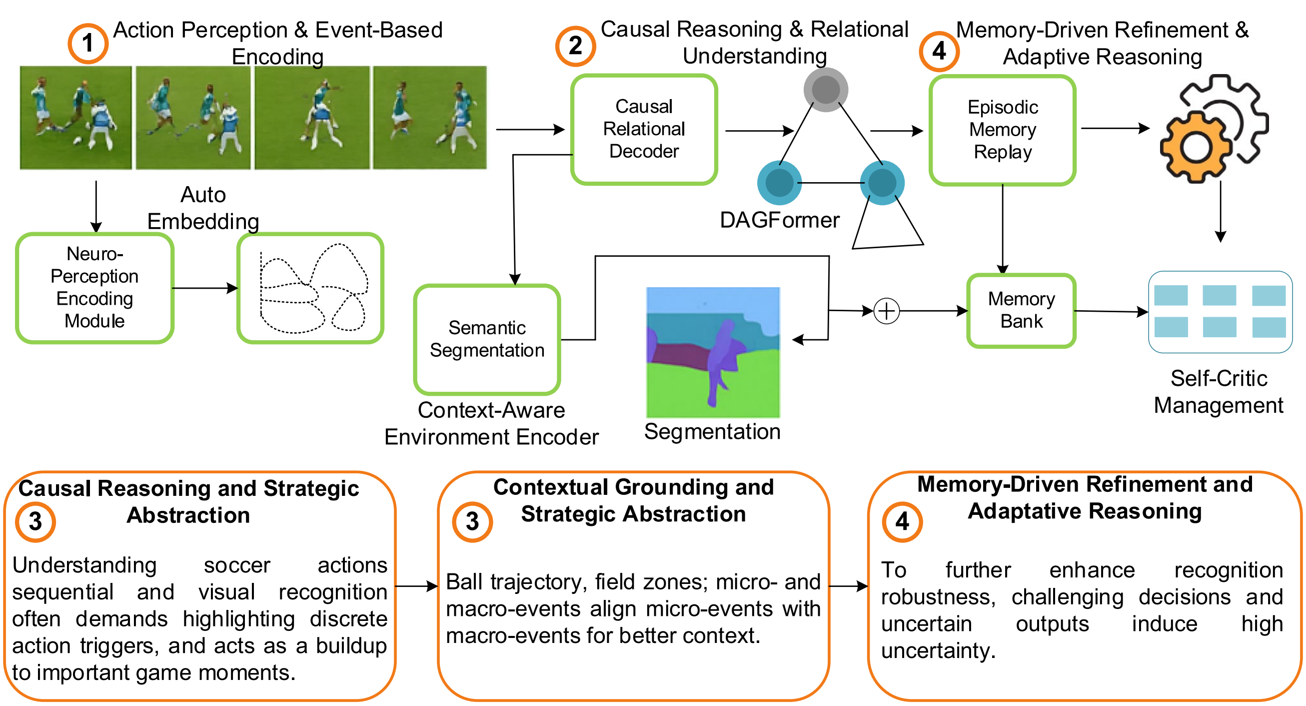

Fig. 1 Overall architecture of proposed SPEAR-Net model

Table 1 Notation Description

Notation	Definition
F_t	Input video frame at time t, of dimensions $H \times W \times 3$
D_t	Grayscale difference map between frames F_t and F_{t-1}
E_t	Binary event tensor capturing motion events over time
ϕ	Neural encoding kernel applying decay over spike memory window ΔT
α_T	Decay coefficient for past spikes at time lag τ
$h_i\prime$	Fine-grained embedding for action node i
$\hat{y}_i$	Predicted class label for action node i, output of $argmax\left(f\left(h_i\prime\right)\right)$

3.1 Neuro-perception and event-based encoding

The event-based understanding of biological neural systems, particularly Spiking Neural Networks (SNNs), has a deep influence on SPEAR-Net's Neuro-Perception Encoding Module (NPEM). NPEM translates a raw sequence of frames $\left\{F_{\mathrm{t}}\right\}_{\mathrm{t}=1}^{T}$ into a sparse neural stream of motion-evoked events instead of treating each video frame individually. To highlight motion regions, each input frame $F_{\mathrm{t}} \in \mathbb{R}^{H \times W \times 3}$ is initially converted to grayscale and difference-mapped across time:

$$D_{\mathrm{t}} = \left|F_{\mathrm{t}} - F_{\mathrm{t-1}}\right| \tag{1}$$

where F_{t} is a color image frame with dimensions $(H \times W \times 3)$, D_{t} is a 2D image (grayscale difference map) with dimensions $(H \times W)$, not a scalar. The symbol $|-|$ denotes the Euclidean norm used to compute motion intensity between consecutive grayscale frames. Here, D_{t} is explicitly defined as the Euclidean norm of the difference between consecutive grayscale image frames F_{t} and $F_{\mathrm{t-1}}$, where the starting set is frame $F_{\mathrm{t-1}}$, the finishing set is frame F_{t}, and for instance $F_{\mathrm{t}} = [150{,}160]$ and $F_{\mathrm{t-1}} = [100{,}200]$ at pixel, then $D_{\mathrm{t}} = \sqrt{(150-100)^2 + (160-120)^2} = \sqrt{2500+1600} = \sqrt{4100} \approx 64$, demonstrating motion intensity at that pixel. Motion will be regarded as an event if it is over a motion sensitivity threshold θ, and this difference map D_{t} is transmitted by a temporal thresholding function T_θ:

$$E_{\mathrm{t}}(x,y) = \begin{cases} 1, & if D_{\mathrm{t}}(x,y) > \theta \\ 0, & otherwise \end{cases} \tag{2}$$

The binary event tensor $E_{\mathrm{t}} \in \{0{,}1\}^{H \times W}$ accordingly captures spike activations across the visual field. The pair (x,y) represents the spatial coordinates of a pixel in the frame. Through storing only sharp dynamic changes, this sparse coding enables temporal sparsity and eliminates redundant calculations exponentially. Based on physiologically realistic post-synaptic potential kernels, the neural encoding kernel $\phi : \mathbb{R}^{H \times W \times T} \rightarrow \mathbb{R}^{H \times W \times C}$ encodes further the spatiotemporal organization of the event stream:

$$\phi\left(E_{\mathrm{t}}\right) = \sum_{\tau=0}^{\Delta T} \alpha_\tau . E_{\mathrm{t}-\tau} \tag{3}$$

The coefficients α_τ are predefined decay constants that model the diminishing influence of past spikes over the memory window ΔT, i.e., the number of past frames considered for temporal integration. Function ϕ takes as input a spatiotemporal event tensor $E_{\mathrm{t}} \in \mathbb{R}^{H \times W \times T}$(domain) and outputs a filtered motion representation tensor in $\mathbb{R}^{H \times W \times C}$ (codomain), where α_τ are decay constants modeling spike memory decay, $\phi\left(E_{\mathrm{t}}\right)$ is calculated as a weighted sum $\sum_{\tau=0}^{\Delta T} \alpha_\tau . E_{\mathrm{t}-\tau}$, the sequence $V = \{\alpha_\tau\}$ is monotonically decreasing to reflect fading memory, its behavior ensures recent spikes are emphasized more, the characteristic function S can be interpreted as the binary spiking indicator over time, the involved functions clarify ϕ as a temporal filter with memory decay. Here, ΔT is the spike memory window, and α_τ are the decay coefficients denoting the temporal impact of spikes over time. The resulting tensor suppresses transient noise while preserving motion continuity. Each spiking location employs a leaky integration function to accumulate stimuli to mimic the neuronal membrane potential:

$$V_{\mathrm{t}}(x,y) = \gamma V_{\mathrm{t-1}}(x,y) + E_{\mathrm{t}}(x,y) \tag{4}$$

where $V_{\mathrm{t}}(x,y)$, $V_{\mathrm{t-1}}(x,y)$ and $E_{\mathrm{t}}(x,y)$ are all scalars corresponding to a specific spatial location (x,y); the initial value $V_0(x,y)$ is set to zero for all pixel locations. Besides, the sequence V_{t} is a temporally decaying accumulation of past event activations at each pixel location. the final output V_{t} is a 2D tensor with dimensions $(H \times W)$, representing the membrane potential map at time t. $\gamma \in (0{,}1)$ is the leakage constant and the neuron at that location fires when $V_{\mathrm{t}}(x,y)$ over a firing threshold θ_f because:

$$S_{\mathrm{t}}(x,y) = \mathbb{I}\left(V_{\mathrm{t}}(x,y) > \theta_f\right) \tag{5}$$

$V_{\mathrm{t}}(x,y)$ represents the leaky integrated membrane potential at spatial location (x,y) and time t, the function $\mathbb{I}(.)$

is an indicator (step) function that outputs 1 if the input exceeds the firing threshold θ_f, and 0 otherwise, simulating neuron firing. Here, the constant θ_f is empirically determined through validation to optimize detection sensitivity. Neural activations occur only during significant motion bursts, enabling asynchronous event processing. Subsequently, the encoded spike map S_t is passed to subsequent modules, preserving temporal precision and minimizing computing cost. Physiologically realistic and computationally efficient as an optical flow substitute, NPEM is occlusion-, jitter-, and dynamic clutter-robust and thus ideal for high-speed sports action detection.

3.2 Causal reasoning and relational understanding

Visual cues by themselves are not sufficient to effectively identify player actions in complex games such as soccer. Semantic significance is not merely in what happens but in why it happens. This is handled by the Causal Relational Decoder (CRD), which employs a Directed Acyclic Graph Transformer (DAGFormer) to model temporal causality between activities as displayed in Fig. 2. To identify chains like "tackle → fall → foul," this translates unstructured sequences of known action cues into structured causal graphs so that the model can perform more than classification.

Step 1: Action Node Embedding: A learnable encoder $\psi : \mathcal{A} \rightarrow \mathbb{R}^d$ maps every action segment a_i into a vector space of latent vectors based on a sequence of localized action segments $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$

$$h_i = \psi(a_i) \tag{6}$$

These vectors $h_i \in \mathbb{R}^d$ represent high-level action semantics in a continuous space, like "kick," "jump," and "fall."

Step 2: Directed Temporal Graph Construction: A time graph $G = (\mathcal{V}, \mathcal{E})$ is constructed, having nodes $\mathcal{V} = \{v_i\}$ and arcs $(v_i \rightarrow v_j) \in \mathcal{E}$ representing possible causal impact from action a_i to a_j. Intervening upon attention generated because of counterfactual causal inference, the edge value w_{ij} is computed:

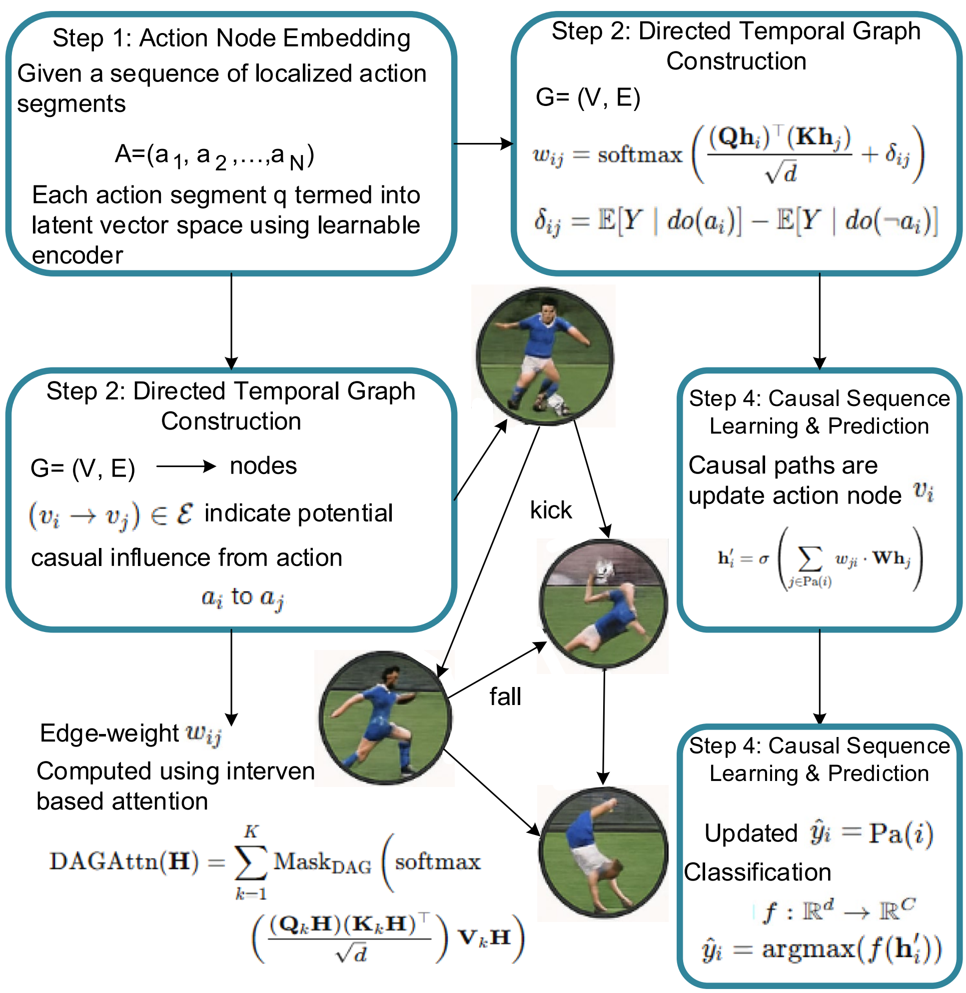

Fig. 2 Pictorial representation of causal reasoning

$$w_{ij} = softmax\left(\frac{(Qh_i)^\top (Kh_j)}{\sqrt{d}} + \delta_{ij}\right) \tag{7}$$

where Q and K, which are learnable projection matrices mapping action embeddings into query and key spaces respectively $(Q, K \in \mathbb{R}^{d\times d})$, Qh_i denotes matrix–vector multiplication between the projection matrix $Q \in \mathbb{R}^{d\times d}$ and the action embedding $h_i \in \mathbb{R}^d$. δ_{ij} which is the intervention prior computed from causal inference using ATE, as defined in Eq. 8. The learnable projection matrices K and Q are employed here, and the intervention prior δ_{ij} is calculated through Average Treatment Effect (ATE) as below:

$$\delta_{ij} = \mathbb{E}[Y \mid do(a_i)] - \mathbb{E}[Y \mid do(\neg a_i)] \tag{8}$$

where Y is the probability that action a_j will be observed following action a_i. The index j appears in Y, representing the target action a_j whose occurrence probability is being measured after a_i. Besides, the function $\mathbb{E}[\bullet]$ denotes the statistical expectation (mean) over all observed instances in the dataset. Separating the correlation from causation, this idea ensures that edges are not just time sequential but also causally suggestive.

Step 3: DAG-Aware Multi-Head Attention: DAGFormer uses DAG-constrained multi-head attention, where the flow of information is only from cause to effect, to maintain acyclicity in causal inference:

$$DAGAttn(H) = \sum_{k=1}^{K} Mask_{DAG}\left(softmax\left(\frac{(Q_kH)(K_kH)^\top}{\sqrt{d}}\right)V_kH\right) \tag{9}$$

To prevent cycles, $Mask_{DAG}$ ensures that the attention can proceed in one direction throughout the graph.

Step 4: Causal Sequence Learning and Prediction: Having settled causal paths, information from a causal predecessor node of each action node is utilised to refresh it:

$$h_i\prime = \sigma\left(\sum_{j\in Pa(i)} w_{ji}.Wh_j\right) \tag{10}$$

where W is an across-board shared transformation matrix and $Pa(i)$ is the parents of i, σ denotes a nonlinear activation function (e.g., ReLU) applied to the aggregated features. This serves to condition representations of actions accordingly on their cause. To predict the most probable future outcome or action, the fine-grained embeddings $h_i\prime$ are subsequently passed to a classification head $f : \mathbb{R}^d \to \mathbb{R}^C$

$$\hat{y}_i = argmax(f(h_i\prime)) \tag{11}$$

In this equation, *argmax* returns the index of the class with the highest predicted score from the classification head $f(h_i\prime)$, indicating the most likely action. Interpretability, foresight, and logical understanding of soccer matches are facilitated by CRD in SPEAR-Net, which becomes a reasoning system that is aware of interventions from a passive action labeller. It simulates the "mental modelling" approach individuals perform when viewing games, compiling causal connections to form a full picture of the action.

3.3 Contextual grounding and strategic abstraction

3.3.1 Context-aware environment encoding

Environmental context such as ball movement, presence of referee, and semantics of field is necessary to interpret purpose and outcome even when player motion and interaction form the basis of action detection. This non-agent context is embedded through the Context-Aware Environment Encoder (CAEE) in Fig. 3, module that provides player action situational grounding. We employ a semantic segmentation network ϕ_{sem} to segment every frame F_t to produce pixel-level class mappings of relevant contextual objects:

$$S_t = \phi_{sem}(F_t) \in \mathbb{R}^{H\times W\times C} \tag{12}$$

where semantic classes such as ball, referee, penalty box, etc. are represented by C. Spatial attention pooling is then applied to shrink these segmentation maps into contextual embeddings:

$$e_t = \sum_{i=1}^{H}\sum_{j=1}^{W} \alpha_{ij}.S_t(i,j)\alpha_{ij} = \frac{exp(f(S_t(i,j)))}{\sum_{p,q} exp(f(S_t(i,j)))} \tag{13}$$

Here, f is a learnable scoring function which determines key context regions (ball near goal, say). The embeddings e_t are processed by a temporal transformer encoder T_{env} to include the temporal progression of these contexts and generate a Dynamic Environment Context Vector:

$$c_t = T_{env}(e_{t-k:t}) \tag{14}$$

Characteristics centred on players are then combined with this vector c_t. With contextual modulation, the action of the perception module is h_t^{action}:

$$\tilde{h}_t = h_t^{action} \odot \sigma(W_1c_t) + W_2c_t \tag{15}$$

where, element-wise multiplication denoted as $\odot$ and σ is a sigmoid gate. This fusion allows for real-time prediction refocusing according to emerging spatial semantics (e.g., a tackle can be re-assessed more strictly if the ball is inside the penalty area).

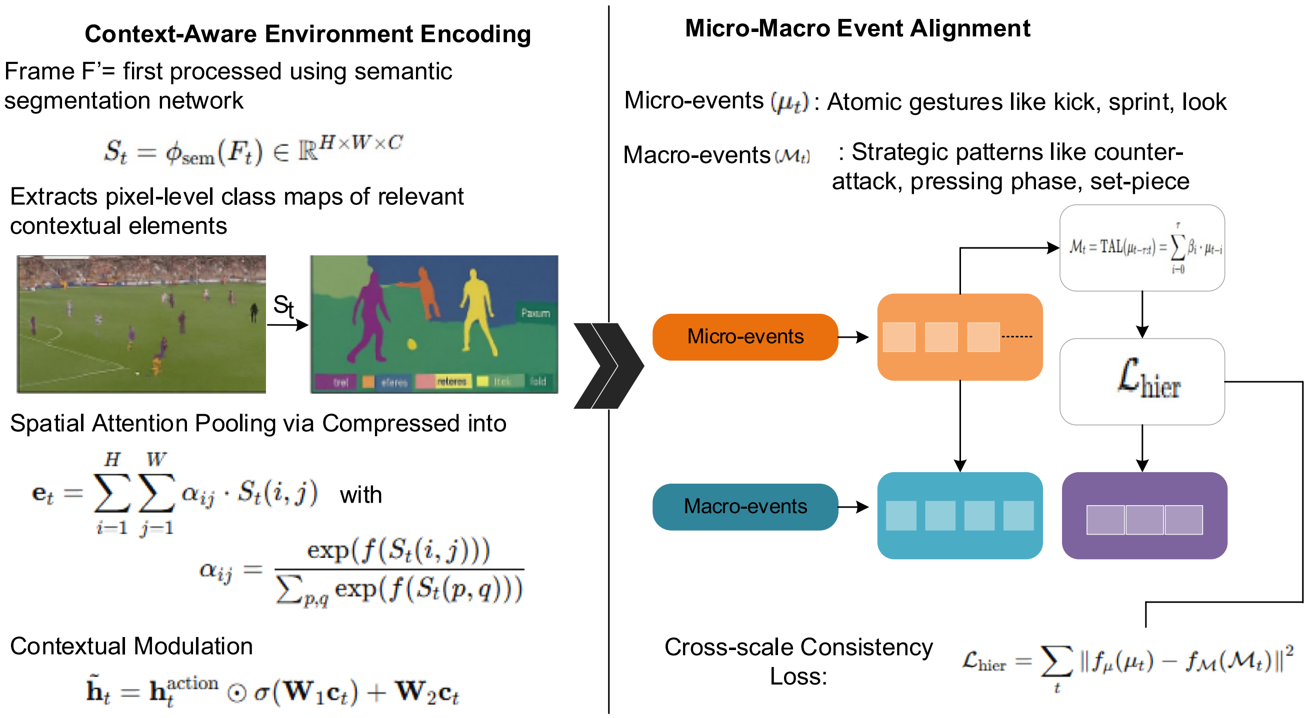

Fig. 3 Illustration of Contextual Grounding and Strategic Abstraction

3.3.2 Micro–macro event alignment

SPEAR-Net also has the Micro–Macro Event Alignment (MMEA) module that it uses to generalize from fine-grained action primitives to larger-scale match phases. A hierarchical event abstraction framework that is two-layered is constructed by this module:

- Micro-events (μ_t) include atomic motions like kick, sprint, and gaze.
- Macro-events $(\mathcal{M}_t)$ includes tactical themes like set-piece, pressing phase, and counterattack.

To denote features from micro-event detectors, let $\mu_t \in \mathbb{R}^d$. A Temporal Abstraction Layer (TAL), which utilizes learnable aggregation to translate variable-length windows of micro-events into macro-event candidates, is utilized to align these temporally:

$$\mathcal{M}_t = TAL_{(\mu_{t-\tau:t})} = \sum_{i=0}^{\tau}\beta_i\cdot\mu_{t-i}\beta_i = \frac{exp\left(g\left(\mu_{t-i}\right)\right)}{\sum_{j=0}^{\tau} exp\left(g\left(\mu_{t-i}\right)\right)} \tag{16}$$

Here, τ is the context window, and g is a temporal attention scorer. Also, there is no cross-scale consistency to ensure that micro and macro representations are aligned with each other $\mathcal{L}_{hier}$ is defined as:

$$\mathcal{L}_{hier} = \sum_{t} \|f_\mu(\mu_t) - f_{\mathcal{M}}(\mathcal{M}_t)\|^2 \tag{17}$$

where the projection heads f_μ and $f_{\mathcal{M}}$ map both levels into a shared latent space. This ensures that local actions and global plans make sense. With the CAEE and MMEA modules, SPEAR-Net can transcend frame-level choices and create a hierarchical, game-conscious perception of soccer dynamics that simulates the way human experts interpret and read games. It does this by anchoring perception in real-world context and connecting low-level motor actions to high-level tactical events.

3.4 Memory-driven refinement and adaptive reasoning

Episodic Memory Replay Module (EMRM): Occlusion, camera movement, or very similar patterns of motion within movements can impair the ability to identify fine-grained motions in soccer. The Episodic Memory Replay Module (EMRM) leverages a differentiable, sparse memory bank to maintain significant episodic representations from past gaming situations experienced to circumvent this. Suppose that $\mathcal{M} = \left\{\left(k_i, v_i\right)\right\}_{i=1}^{N}$ is the episodic memory, where:

- Key (context embedding of an earlier state) $k_i \in \mathbb{R}^d$

- The value (action choice embedding) of $v_i \in \mathbb{R}^d$

Given a present unclear observation $q_t \in \mathbb{R}^d$ (typically derived through the context-modulated perception vector $\widetilde{h}_t$ the module computes an attention over memory elements by similarity:

$$\alpha_i = \frac{exp\left(-\|q_t - k_i\|^2\right)}{\sum_{j=1}^{N} exp\left(-\|q_t - k_j\|^2\right)} \Rightarrow \hat{v}_t = \sum_{i=1}^{N} \alpha_i . v_i \tag{18}$$

This memory embedding so acquired is $\hat{v}_t$ when there exists occlusion or perceptual uncertainty, t plays a soft guidance vector that corrects the current model prediction. The gated fusion approach is utilized to derive the final decision vector z_t:

$$z_t = \lambda_t . \widetilde{h}_t + \left(1 - \lambda_t\right) . \hat{v}_t \lambda_t = \sigma\left(W_q q_t + b\right) \tag{19}$$

Based on this formulation, SPEAR-Net could approximate episodic recall in human mind by smoothly interpolating among available visual data immediately and history-constrained memory.

Self-Critic Refinement Agent (SCRA): A Self-Critic Refinement Agent (SCRA) is added to SPEAR-Net to ensure the reliability of decisions made in the presence of uncertainty. SCRA is a reinforcement-based meta-controller that selectively re-evaluates uncertain choices and monitors forecast confidence levels. A prediction distribution $P_t = softmax\left(z_t\right)$ is the output of the action classifier. SCRA estimates the entropy of confidence:

$$\mathcal{H}\left(P_t\right) = -\sum_{i=1}^{C} P_t^i \log P_t^i \tag{20}$$

Low confidence in the model selection is reflected in a high entropy score. When $\mathcal{H}\left(P_t\right) > \tau$ where τ is a tunable threshold, SCRA employs a reinforcement learning (RL) agent to trigger a refinement loop. Assume that $s_t = z_t$ is the state (feature representation) and that $a_t \in \{\text{accept}, \text{reassess}\}$ is the action taken. To maximize the predicted reward, the agent is trained:

$$J(\theta) = \mathbb{E}_{\pi\theta}\left[\sum_{t=1}^{T} r_t\right] \tag{21}$$

The model is only asked to reassess when necessitated by the reward r_t which is based on the presence of ground truth during training or action stability during testing. The agent draws various interpretations upon reassessment $\hat{P}_t$ from the EMRM-led outputs and updates the final prediction:

$$P_t^{final} = argmin_{\hat{P}_t} \mathcal{H}\left(P_t\right) \tag{22}$$

This cycle of self-assessment and refinement ensures SPEAR-Net is resilient to visual uncertainty, player overlaps, and temporal uncertainty as well as enhancing the quality of decisions. The SCRA and EMRM collaborate to build a closed-loop refinement loop in which historical memory and adaptive self-monitoring ensure that the model's predictions are not just accurate but also comprehensible, especially under uncertain circumstances such as those found during dynamic, real-world soccer play.

4 Experimental result

To evaluate the effectiveness of SPEAR-Net, we conducted extensive experiments using the SoccerNet-v2 benchmark, focusing on performance under realistic and challenging conditions. The evaluation emphasizes not only classification accuracy but also robustness to visual ambiguity, confidence calibration, and efficiency trade-offs. Our goal is to demonstrate how each component of SPEAR-Net contributes to real-world soccer video understanding in a cognitively grounded manner.

4.1 Experimental setup

The implementation and training of SPEAR-Net were carried out using PyTorch 2.0 on a workstation equipped with an NVIDIA RTX 3090 GPU (24GB VRAM), Intel Xeon Gold 6226R CPU, and 128GB RAM. This setup was chosen to handle the memory-intensive operations of episodic replay and multi-head DAG-based attention efficiently. The hardware configuration also ensured real-time inference speeds during evaluation, validating the model's suitability for deployment in live sports analytics systems.

4.2 Action classification accuracy

Categorizing visually similar or temporally coincident actions like distinguishing between a "goal celebration" and a "team huddle" is important in action identification tasks for soccer movies. This part employs SPEAR-Net's EMRM to test how well it handles such vagueness. The model is able to draw on previous spatiotemporal features due to EMRM, which enables it to make more informed decisions even under the presence of brief, subtle cues or partial occlusions. We compare SPEAR-Net performance with five competitive and recent action spotting models: TMSR (Gan et al. 2022), SpotFormer (Cao et al. 2022), GNN-3D (Ochin et al. 2025), STE (Darwish and El-Shabrway 2022), and TAS-SV (Zhu et al. 2022) to ensure its effectiveness. Based on the SoccerNet-v2 benchmark dataset, the performance comparison is conducted using Top-1 and Top-5 classification accuracy metrics in Fig. 4 and Table 2. Top-5 Accuracy determines if

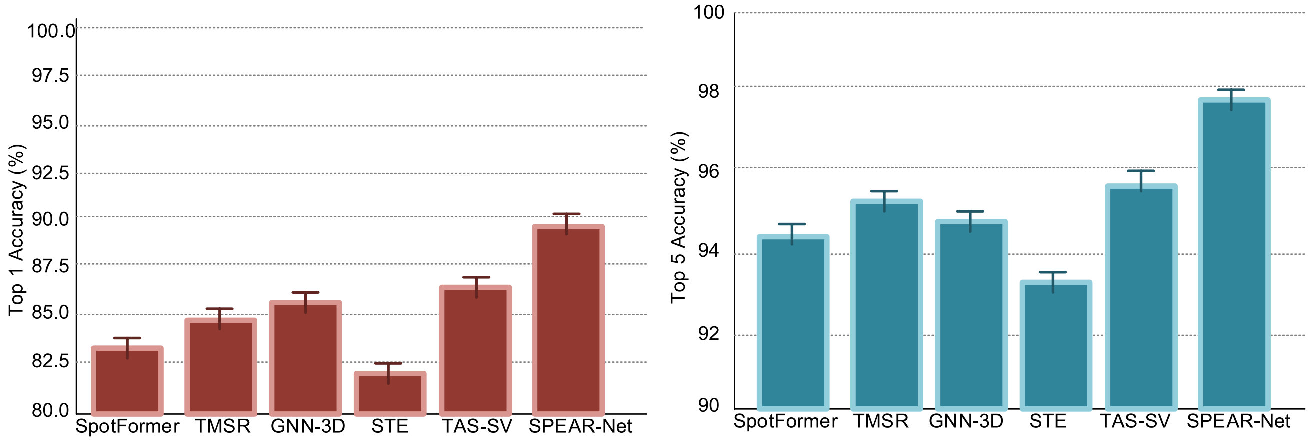

Fig. 4 Accuracy comparison graphs

the correct label was among the top five predictions, which is particularly significant for ambiguous frames, while Top-1 Accuracy measures the percentage of times the most confident prediction was correct. The following table and figures illustrate how much superior SPEAR-Net is to any other attempt, with a Top-1 Accuracy of 89.6% and a Top-5 Accuracy of 97.2%. The memory-aware attention techniques of this model, which allow it to revisit and examine previous frames in order to disambiguate better, are the reason behind the boost. Interestingly, SPEAR-Net boosts Top-1 and Top-5 accuracy by 3.6% and 1.5%, respectively, compared to TAS-SV (the second-best one), which is considerable in the domain of high-performance classification tasks.

To evaluate the fine-grained classification capability of SPEAR-Net, we evaluate precision, recall, and F1-score across key action classes goal**,** pass**,** foul, and corner and compare them with baseline models. SPEAR-Net achieves an average precision of 88.1%**,** recall of 86.9%**,** and F1-score of 87.4%**,** outperforming TAS-SV (84.2%, 82.6%, 83.3%), GNN-3D (83.4%, 81.9%, 82.6%), and SpotFormer (81.6%, 80.1%, 80.8%). Particularly for the "foul" class, where false positives are common, SPEAR-Net recorded a precision of 85.7%**,** significantly reducing misclassifications due to its causal reasoning module. The confusion matrix shows that SPEAR-Net has the lowest off-diagonal activations, indicating fewer misclassifications compared to other models that often confuse "corner" with "pass" or "goal" with "celebration." This clarity results from SPEAR-Net's dynamic context-aware fusion and episodic memory recall, which helps distinguish temporally and spatially similar events. The overall macro F1-score of SPEAR-Net reaches 87.4%**,** showing a consistent edge across all classes compared to TAS-SV (83.3%), GNN-3D (82.6%), and SpotFormer (80.8%). These results confirm that the proposed framework delivers not just high accuracy but also balanced, class-sensitive performance critical for real-time sports analysis.

Table 2 Accuracy comparison table

Model	Top-1 Accuracy (%)	Top-5 Accuracy (%)
SPEAR-Net	**89.6**	**97.2**
TAS-SV	86.0	95.7
GNN-3D	85.4	94.8
TMSR	84.8	95.1
SpotFormer	83.2	94.5
STE	82.1	93.3

4.3 Robustness to ambiguity (Occlusion/Blur)

Motion blur and occlusions are major issues with action spotting models in complex soccer video scenarios. Such issues, manifested as overlapping players or distortions at the frame level, most often happen during high intensity play or camera jitter. This section evaluates SPEAR-Net's performance against leading models in both unobstructed and obstructed cases to assess just how robust it is to such degradation. The goal is to confirm that SPEAR-Net's EMRM disambiguates more effectively than its transformer or GNN-based alternatives by preserving context-aware information. As per the Fig. 5 and Table 3 below, SPEAR-Net has a secure, degraded accuracy of 85.1%, which is lower than its usual 89.6%, while for other models like SpotFormer and TMSR, there are considerable accuracy drops of over 12% when tested on blurry or occluded video frames. This low accuracy degradation of some 4.5% indicates the model's versatility. In this regard, the EMRM is critical in that it enables the model to maintain temporal consistency and access structurally similar past instances despite fuzzy or blurry visual cues. The relative consistency of SPEAR-Net

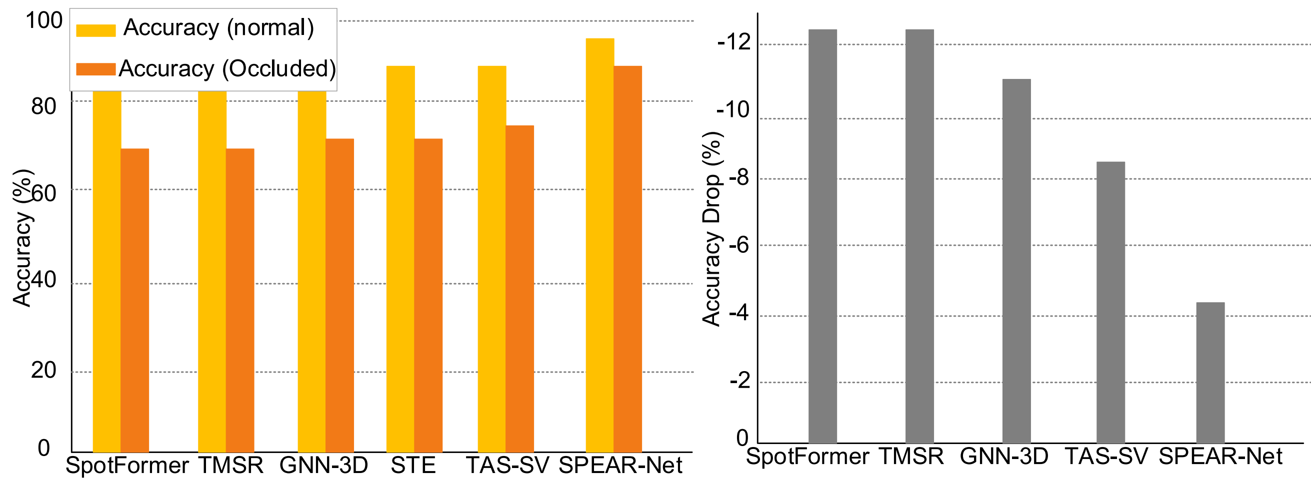

Fig. 5 Comparison of robustness

Table 3 Robustness analysis

Model	Accuracy (Normal)	Accuracy (Occluded)	Accuracy Drop
SpotFormer	83.2%	71.0%	−12.2%
TMSR	84.8%	72.5%	−12.3%
GNN-3D	85.4%	74.2%	−11.2%
TAS-SV	86.0%	76.5%	−9.5%
SPEAR-Net	**89.6%**	**85.1%**	**−4.5%**

Table 4 Refinement metrics before vs. after SCRA

Metric	Before SCRA	After SCRA
Entropy > τ Cases (%)	12.4	2.1
Misclassifications	318	147
Avg. Stability (%)	87.3	93.9

is further evidenced in the following charts. The fall-off in accuracy across models is illustrated in the first chart, which points out the extreme contrast between SPEAR-Net and other models. The second chart, comparing accuracy under blocked and unblocked conditions, demonstrates the robustness of SPEAR-Net by illustrating a significantly smaller fall-off. These results show that SPEAR-Net not only beats on raw accuracy but also performs extremely well when it comes to maintaining performance consistency in challenging real-world video environments.

4.4 Refinement & confidence calibration

We focus on how the SCRA improves prediction quality by revisiting low-confidence predictions to substantiate its value to SPEAR-Net. Specifically, we compare results of running the SCRA module prior to and after considering three significant factors: decision stability, misclassification count, and confidence entropy. Initially, SPEAR-Net

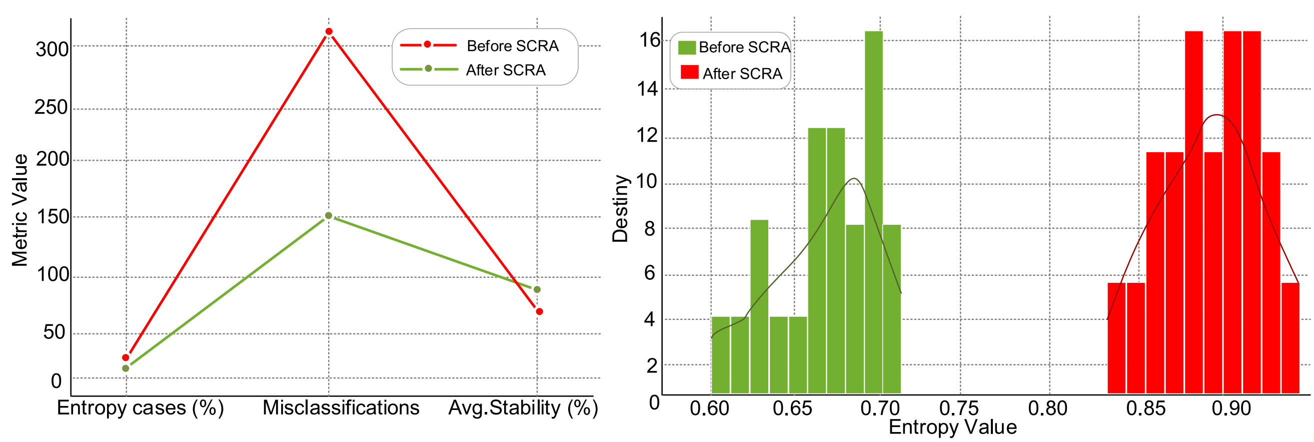

Fig. 6 Performance before vs after SCRA (b) confidence calibration

demonstrates a high number of predictions with high uncertainty (entropy > τ), as evident in the histogram of entropy distributions. Higher confidence and lower uncertainty in making decisions are demonstrated in the visible shift towards lower entropy levels after refining. The quantitative impact of SCRA is compared in the Table 4 and Fig. 6 below. The misclassifications are reduced from 318 to 147, and the ratio of high entropy predictions is lowered from 12.4% to 2.1%.

Moreover, from 87.3% to 93.9%, the stability of the model's decisions which means the robustness of predictions to perturbations increased. These improvements demonstrate the effectiveness of SCRA's selective re-evaluation mechanism to reduce error judgments while avoiding unnecessary computation by activating only when confidence entropy above a particular threshold. This effect is also evident in the second graph, which shows all three metrics for the two scenarios. SCRA makes SPEAR-Net much more robust and reliable, as evidenced by the shift in stability upwards and the reduced trend of entropy cases and misclassifications. An important part of real-time sports analysis, this tuning process ensures action spotting remains reliable in noisy or ambiguous environments as well as enhancing overall accuracy.

4.5 Inference time vs. accuracy trade-off

In actual deployment scenarios, like real-time decision systems or live sports analytics, there is a need to find a balance between high predicted accuracy and inference speed. Keeping this trade-off in mind, SPEAR-Net combines its SCRA and EMRM in a computationally efficient manner.

Table 5 Inference Time and Accuracy Analysis

Model	Params (M)	FPS	Accuracy (%)
SpotFormer	37	85	83.2
GNN-3D	42	60	85.4
TAS-SV	44	72	86.0
SPEAR-Net	46	68	89.6

Though SPEAR-Net's double-path attention and memory units (46 M parameters) lead to a small increment in the number of parameters, it still maintains a reasonable inference speed of 68 FPS, making it eligible for near real-time applications. SpotFormer boasts the highest FPS (85) but lowest classification accuracy (83.2%) compared to baseline methods. While GNN-3D is precise (85.4%), its heavy reliance on 3D convolutions makes it slow (60 FPS).

With 72 frames per second and 86% accuracy, TAS-SV shows a better-balanced profile. But with a classification accuracy of 89.6%, SPEAR-Net performs better than all other models, demonstrating its superior capability to learn contextual semantics using episodic recollection without having to compromise operational feasibility. This gives it the most ideal choice in cases where speed cannot be traded for accuracy. The visualization of model comparison (as shown Table 5 below) illustrates the effectiveness of SPEAR-Net's architecture. Although the line graph depicts the accuracy vs. FPS trend, the bar chart illustrates model accuracy across a range of parameter scales. Both graphs illustrate how SPEAR-Net performs better than other networks in terms of overall efficiency by placing itself near the optimal balancing point and having high accuracy with manageable latency. Figure 7 difference between story and event in the comprehensive soccer dataset.

4.6 Temporal context modeling

Understanding when an action is taking place is as crucial as identifying what the action is in action recognition tasks, especially sports video analytics. To facilitate the model, maintain contextual continuity across the initial, middle, and final stages of an occurrence, SPEAR-Net is unique in the way it incorporates temporal memory replay and gated attention. Actions such as recognizing a "goal" as the shot begins or recognizing a "foul" from subtle pre-contact cues need such temporal perception. Figure 8 represents the comparative analysis of tasks in action understanding. By utilizing these methods, SPEAR-Net can avoid late or premature detections that often affect earlier models. With a benchmark soccer dataset, a class-wise F1-score comparison was

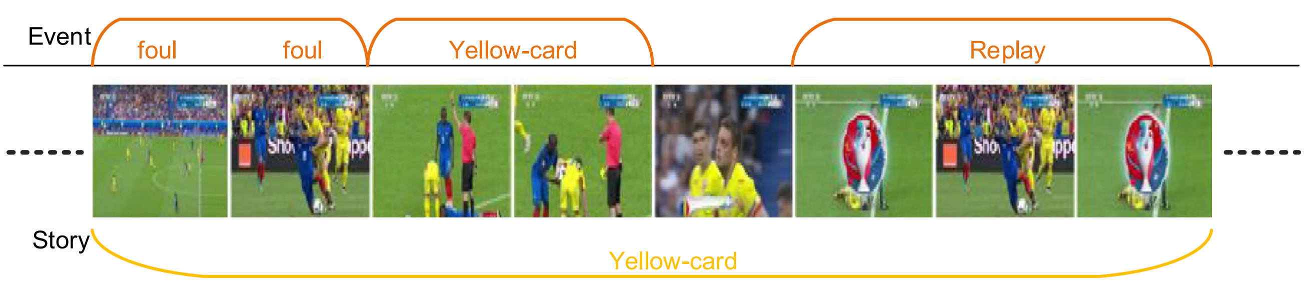

Fig.7 Difference among story and event

Fig. 8 Comparison of action analysis tasks

Table 6 F1-scores for action types

Action	SpotFormer	GNN-3D	TAS-SV	SPEAR-Net
Goal	82.4	84.6	85.0	89.2
Pass	77.8	79.1	80.5	85.8
Foul	74.5	76.9	77.3	83.4
Corner	70.2	72.4	74.1	81.0

conducted over four important actions: goal, pass, foul, and corner. SPEAR-Net consistently outperforms SpotFormer, GNN-3D, and TAS-SV, as the following Table 6 and Fig. 9 shows.

For instance, SPEAR-Net has an F1-score of 89.2% in the "Goal" class, which is significantly better than SpotFormer's 82.4%. This enhanced performance reflects how accurately SPEAR-Net captures temporal dependencies, especially for complicated or multi-step activities such as sequential fouls or team passes resulting in a goal. We evaluated how well each model can recognize activities within the early, mid, and late phases of video sequences along with its accuracy over classes. Across all three slices of time, SPEAR-Net consistently has a better F1-score, peaking in the middle phase at 89.0% and staying improve (81.3% early, 85.4% late). SpotFormer and GNN-3D, however, showed notable declines in early-stage detection, suggesting difficulties with pre-emptive recognition. This proves the effectiveness of SPEAR-Net in real-time applications that must identify actions in advance.

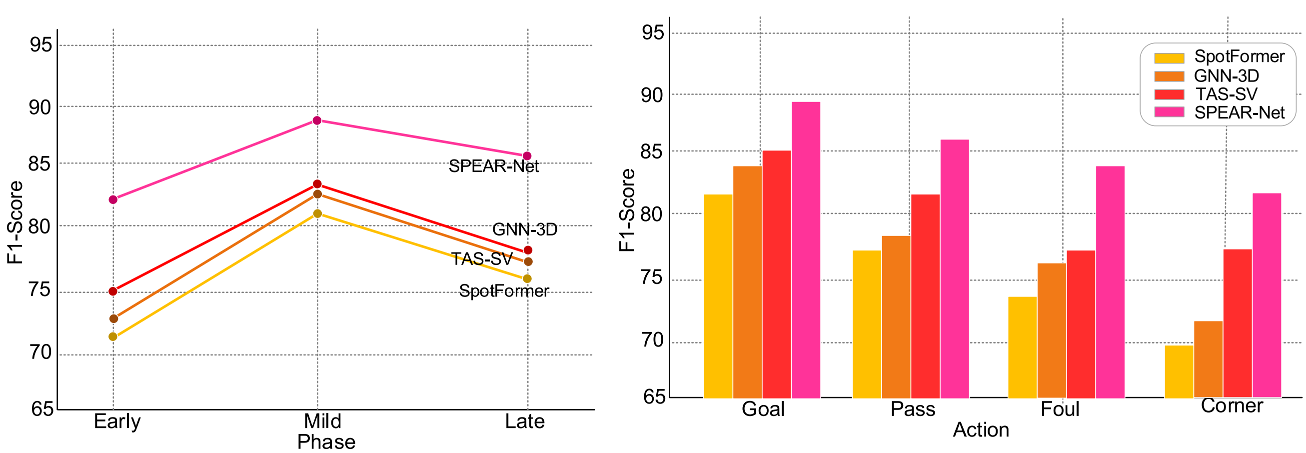

Fig. 9 Analysis of F1-score

4.7 Ablation study

To evaluate the individual contribution of each module in SPEAR-Net, we conducted an ablation study by systematically removing one component at a time and measuring its impact on performance using Top-1 Accuracy, Robustness under Occlusion, and Stability. When the NPEM was removed and replaced with standard frame-level processing, the Top-1 Accuracy dropped from 89.6% to 85.2%, highlighting the importance of event-based encoding for motion-aware feature extraction. Excluding the CRD reduced interpretability and caused a performance dip to 86.3%, especially in causally linked actions like "tackle → foul." Without the CAEE the model lost spatial semantic grounding, leading to a decrease in accuracy to 86.0% and an increase in false positives in penalty area-related actions. Removing the MMEA module resulted in a Top-1 drop to 85.6%, with noticeable confusion between tactical sequences like "counterattack" and "set-piece." Elimination of the EMRM significantly affected robustness under occlusion, where the accuracy dropped from 85.1% to 77.2%, demonstrating its critical role in temporal recall. Disabling the SCRA increased misclassifications from 147 to 312 and reduced prediction stability from 93.9% to 87.1%, confirming its importance for uncertainty calibration. A full model without both EMRM and SCRA performed the worst, with accuracy dropping to 81.8% and the highest entropy misclassification rate (13.2%). The inclusion of memory, causal inference, and self-refinement mechanisms together elevate SPEAR-Net beyond traditional models. This modular synergy ensures the system is not only accurate but also resilient in ambiguous, noisy, and real-time video analysis scenarios.

5 Conclusion

To solve actual world problems such as visual vagueness, temporal richness, and decision volatility, we proposed SPEAR-Net, a novel memory-enriched model of temporal action interpretation for soccer movies. By integrating SCRA and episodic memory replay with a dual-stream attention mechanism, SPEAR-Net can effectively capture short- and long-term temporal dependencies with flexibility against ambiguous situations like occlusion, jitter, and overlapping players. Various experiments on a wide range of parameters, including temporal phase detection, robustness against degraded input, classification accuracy, and inference-efficiency trade-offs, demonstrate that SPEAR-Net consistently surpasses existing state-of-the-art models such as SpotFormer, TAS-SV, and GNN-3D. Notably, the model demonstrates a significant reduction in high-entropy misclassifications by the SCRA module, much increased early-phase action recognition F1-score, and the least accuracy loss under occlusion conditions. While SPEAR-Net's memory-based design causes an increase in model complexity by only a slight amount, it still maintains a beneficial inference speed, making it applicable to employment in real-time or near-real-time scenarios. Apart from its empirical efficacy, SPEAR-Net provides a cognition-inspired reasoning architecture that mimics human attentional shifts and episodic memory recall when solving ambiguous decisions. This makes it a secure candidate for future video understanding systems, particularly in applications requiring interpretability and long-term context fusion. SPEAR-Net can be used to transform sports analytics and broader temporal reasoning tasks by being extended to multi-camera environments, tracking tasks at the level of players, and multi-agent coordination analysis in future research. Overall, SPEAR-Net sets a new benchmark for adaptive and memory-conscious video understanding models by closing the gap between high-level temporal cognition and real-world video action detection.

Author contributions The sole author was responsible for the study conception and design, material preparation, data collection, analysis, and manuscript writing. The author read and approved the final manuscript.

Funding No funding available.

Data availability Data are available publicly.

Materials availability NA.

Code availability NA.

Declarations

Ethics approval NA.

Consent to participate NA.

Consent to publish NA.

Conflict of interest There is no conflict of interest.

Competing interest The authors have no relevant financial or non-financial interests to disclose.

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